



Date: Aug 28, 2025

Math Section 1.1

Practice Test 2

Name:

Class of:

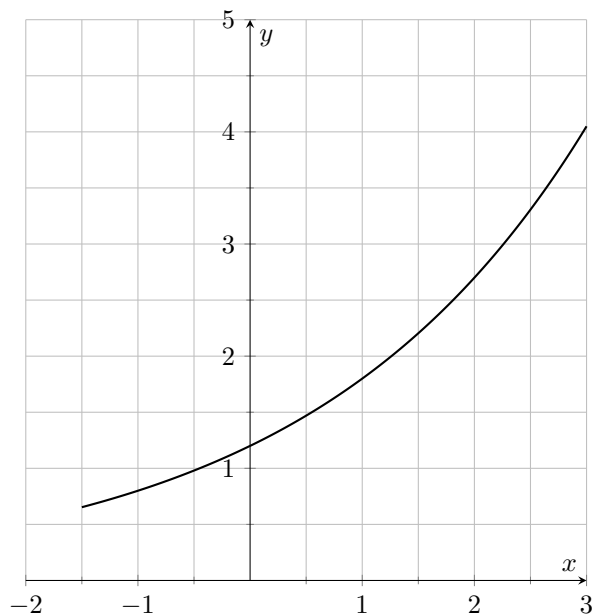
Teacher:

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Solutions

1. Graph Interpretation — Identify a and b

Question. Based on the graph, which of the following statements are true? Select ALL that apply.



1. $a > 0$
2. $a < 0$
3. $b > 1$
4. $0 < b < 1$

Solution (answer: $\boxed{A, C}$). For $f(x) = ab^x$, $a = f(0)$. The graph is above the x -axis at $x = 0 \Rightarrow a > 0$ (A true, B false). The curve increases as x increases (no leveling to 0) $\Rightarrow b > 1$ (C true, D false).

2. Table Interpretation — Determine a and b

Question. Which of the following statements about the function described by the table are true? Select ALL that apply.

x	$f(x)$
0	5
2	45
5	1215

1. This is an exponential decay function.
2. The initial value is positive.
3. There is an asymptote at $y = 0$.
4. There is not enough information to determine whether any of the above statements are true.

Solution (answers: $\boxed{B, C}$). Equal-step ratios: $\frac{f(2)}{f(0)} = \frac{45}{5} = 9 = b^2 \Rightarrow b = 3$; $\frac{f(5)}{f(2)} = \frac{1215}{45} = 27 = b^3 \Rightarrow b = 3$ (checks). Thus $b = 3 > 1$ (growth, so A false). Initial value $a = f(0) = 5 > 0$ (B true). Any ab^x with $b > 0$ has horizontal asymptote $y = 0$ (C true). D is false.

3. Solve an Exponential Equation

Question. What is the value of t in the equation

$$5^{2t-3} = 25^{t+1}$$

1. 2
2. 3
3. 4
4. 5

Solution (answer: D).

1. Rewrite 25 as a power of 5: $25 = 5^2$.
2. Then

$$25^{t+1} = (5^2)^{t+1} = 5^{2(t+1)} = 5^{2t+2}.$$

3. Bases match and are $> 0 \neq 1$, so set exponents equal:

$$3t - 3 = 2t + 2 \implies t = 5.$$

4. *Check:* LHS $= 5^{3(5)-3} = 5^{12}$, RHS $= 25^6 = (5^2)^6 = 5^{12}$

4. Percent Depreciation from a Base

Question. The value $v(t)$ of a car depreciates according to $v(t) = P(0.78)^t$. By what percentage does the value of the car depreciate each year?

1. 0.22%
2. 0.78%
3. 22%
4. 78%

Solution (answer: C). Write $0.78 = 1 - r \Rightarrow r = 1 - 0.78 = 0.22 = 22\%$. Depreciation is 22% per year.

5. Equivalent Exponential Forms

Question. $Q(t) = 7(1.6)^{4t}$. Which equations are equivalent? Select ALL that apply.

$$\text{A } 7(3.2)^t \quad \text{B } 7[(1.6)^t]^4 \quad \text{C } 7(1.6)(1.6)^t \quad \text{D } 7((1.6)^4)^t$$

Solution (answers: B, D). Use $a^{4t} = (a^t)^4 = (a^4)^t$:

$$7(1.6)^{4t} = 7[(1.6)^t]^4 = 7((1.6)^4)^t.$$

Compute $(1.6)^4 = 6.5536$, so $Q(t) = 7(6.5536)^t$. A is invalid (doubles the base instead of applying exponent rules). C equals $7(1.6)^{t+1}$ (changes the exponent).

6. Half-Life Reasoning (Free Response)

Question. A certain antibiotic has a half-life of 6 hours. The nurse says the next dose may be taken in 24 hours, but the patient worries that some of the original dose will still remain. Explain why both can be true.

Solution (model). Exponential decay with half-life 6 h:

$$A(t) = A_0 \left(\frac{1}{2}\right)^{t/6}.$$

Since $A(t) > 0$ for all t and $A(t) \rightarrow 0^+$ as $t \rightarrow \infty$, a small amount always remains (horizontal asymptote $y = 0$). After 24 h there are $24/6 = 4$ half-lives:

$$A(24) = A_0 \left(\frac{1}{2}\right)^4 = \frac{A_0}{16} = 6.25\% \text{ remaining.}$$

This residue is small and keeps halving, so re-dosing at 24 h is considered safe.